



JavaScript: A Crash Course

Part II: Browser-Specific Features

Originals of Slides and Source Code for Examples:
<http://courses.coreservlets.com/Course-Materials/ajax.html>

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Taught by the author of *Core Servlets and JSP*, *More Servlets and JSP*, and this tutorial. Available at public venues, or customized versions can be held on-site at your organization.

- Courses developed and taught by Marty Hall
 - Java 5, Java 6, intermediate/beginning servlets/JSP, advanced servlets/JSP, Struts, JSF, Ajax, GWT, custom mix of topics
- Courses developed and taught by coreservlets.com experts (edited by Marty)
 - Spring, Hibernate, EJB3, Ruby/Rails

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Topics in This Section

- **XML**
 - Getting document
 - Document, Element, and Node classes
- **HTML**
 - HTMLDocument and HTML_ELEMENT classes
 - Specialized HTML_ELEMENT classes
 - The Window class
- **Event Handling**
 - General event-handling attributes
 - Element-specific event-handling attributes



Intro

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Browser-Specific Classes

- **Previous lecture**
 - Described core JavaScript syntax and objects
- **This lecture**
 - Describes JavaScript specifically for browser applications
- **Reminder re references**
 - Online
 - <http://www.w3schools.com/js/>
 - http://www.w3schools.com/html/dom/dom_reference.asp
 - http://www.devguru.com/technologies/ecmascript/QuickRef/javascript_intro.html
 - Book
 - *JavaScript, the Definitive Guide* by David Flanagan



XML

Overview

- **API is very similar to Java DOM API**
 - If you know Java `org.w3c.dom` classes and methods, they are almost the same in JavaScript
- **Document class**
 - Represents top-level document
 - Also a specialized version representing the HTML page
- **Element class**
 - Represents XML/HTML element
 - Inherits Node methods plus has some extras
- **Node class**
 - Represents node in XML tree
 - Element is main node type, but there are also text nodes, CDATA notes, and a few others
 - Most Element methods inherited from here

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Getting Document

- **Getting general XML document: in real life**
 - You get XML document as the result of an Ajax request
 - `var xmlDocument = request.responseXML;`
- **Getting XML document: for practicing and learning**
 - Can use *Firefox-specific* way of parsing a String representing an XML document
 - `String xmlString = "<blah>...</blah>";`
`var parser = new DOMParser();`
`var xmlDocument =`
`parser.parseFromString(xmlString, "application/xml");`
- **Getting XML document representing HTML page**
 - Use builtin "document" variable
 - Many special features apply
 - `getDocumentById` method, `innerHTML` property for elements, forms and anchors and images properties, case-insensitivity for `getElementsByTagName`, lots more specific to HTML
 - http://www.w3schools.com/html/dom/dom_obj_document.asp

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Syntax Summary

- **Document class**
 - Properties
 - documentElement
 - Methods
 - getElementById (HTML only), getElementsByTagName
- **Element class**
 - Methods
 - getAttribute, getElementsByTagName, hasAttribute
- **Node**
 - Properties
 - attributes, childNodes, firstChild, lastChild, nextSibling, nodeName, nodeType, nodeValue, parentNode, previousSibling
 - Methods
 - hasAttributes, hasChildNodes, normalize

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The Document Class

- **documentElement property**
 - The root Element of the document
- **getElementById method**
 - Returns the Element with the specified ID.
 - For HTML documents only!
 - Refers to attribute that the DTD defines as an "id attribute", not necessarily named "id". Does not match attributes named "id" in regular XML docs.
 - Is supposed to be case-sensitive, but IE is case-insensitive
- **getElementsByTagName**
 - Returns an array of Elements that have that tag name
 - Can use "*" for all Elements in document
 - Unsupported in IE 5
 - Is case-sensitive for regular XML documents
 - Is case-insensitive for HTML documents
 - Even when using XHTML

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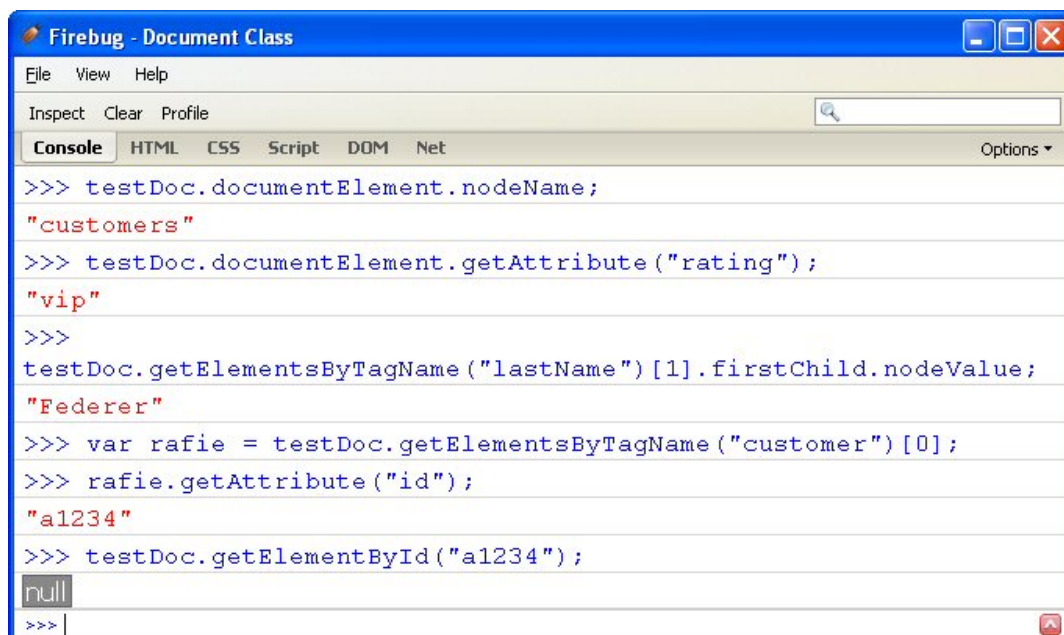
The Document Class: Examples (Code)

```
function getXmlDoc(xmlString) {
    var parser = new DOMParser();
    var xmlDocument =
        parser.parseFromString(xmlString, "application/xml");
    return(xmlDocument);
}
var test =
    "<customers rating='vip'>" +
    "  <customer id='a1234'>" +
    "    <firstName>Rafael</firstName>" +
    "    <lastName>Nadal</lastName>" +
    "  </customer>" +
    "  <customer id='a1235'>" +
    "    <firstName>Roger</firstName>" +
    "    <lastName>Federer</lastName>" +
    "  </customer>" +
    "</customers>";
var testDoc = getXmlDoc(test);
```

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The Document Class: Examples (Results)



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The Element Class

- **getAttribute**
 - Gets value of designated attribute.
 - E.g., if element refers to `<foo bar="a" baz="b">...</foo>`, `element.getAttribute("baz")` returns "b"
- **getElementsByTagName**
 - Returns an array of subelements that have this tag name
 - Subelements can be arbitrarily nested
- **hasAttribute**
 - Tests if element has attribute of given name
- **Also inherits from Node class**
 - See next slides
 - All Elements are Nodes, but not vice versa

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The Node Class: Properties

- **attributes**
 - An array of the attribute names of the Node
- **childNodes**
 - An array of direct child nodes. 0-length if no children.
- **firstChild, lastChild, parentNode**
 - Specific child nodes. Parent node (null for top element).
- **nextSibling, previousSibling**
 - Related children of the parent node
- **nodeName**
 - For Element nodes, the XML element name
- **nodeType**
 - `Node.ELEMENT_NODE`, `Node.TEXT_NODE`, `Node.CDATA_SECTION_NODE`, and a few other options
- **nodeValue**
 - For Text nodes, the body content.
 - Call `normalize` first. See next slide.

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The Node Class: Methods

- **hasAttributes**
 - Does this Node have any attributes at all?
- **hasChildNodes**
 - Does this Node have any children at all?
- **normalize**
 - Merges empty text nodes and removes empty ones.
 - Important if element has body content that spans multiple lines. Body content is `nodeValue` of the child text node.
 - `// Element is <foo>bar</foo>`
`var element = someElement();`
`element.normalize();`
`var bodyContent = element.firstChild.nodeValue;`
`// bodyContent is now "bar"`

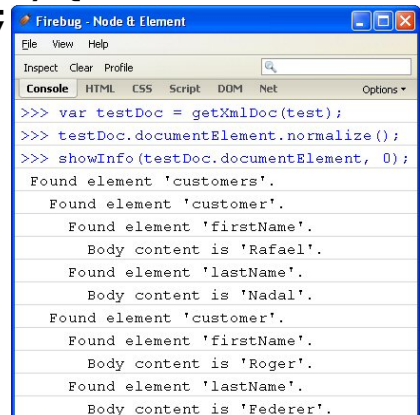
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Node and Element: Example

```
function showInfo(node, indent) {
    if (node.nodeType == Node.TEXT_NODE) {
        console.log("%s Body content is '%s'." ,
            spaces(indent), node.nodeValue);
    } else if (node.nodeType == Node.ELEMENT_NODE) {
        console.log("%s Found element '%s'." ,
            spaces(indent), node.nodeName);
        var children = node.childNodes;
        for(var i=0; i<children.length; i++) {
            showInfo(children[i], indent+1);
        }
    }
}

function spaces(n) {
    var indentString = "  ";
    var result = "";
    for(var i=0; i<n; i++) {
        result =
            result.concat(indentString);
    }
    return(result);
}
```



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Java >>> hm



HTML and Browser Capabilities

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HTMLDocument: Properties

- **Obtain with predefined 'document' variable**
 - Specialized subclass of Document class discussed earlier
- **Properties**
 - title, domain, URL
 - Info about the document. URL is same as `window.location.href` unless redirection occurs
 - body
 - The body element
 - anchors, applets, forms, images, links
 - Arrays of subelements, in the order they appear in the document. Usually better to find elements by ids.
 - cookie, lastModified, referrer
 - In Ajax, it is usually better to manipulate these on server
 - blah
 - Element that has `name="blah"` (first one if repeated)

HTMLDocument: Methods

- **write, writeln**
 - Dynamically insert text into document
 - Used from <script> tag that has body content
 - Not used by Ajax response handlers
 - Use HTMLElement.innerHTML property instead
- **getElementsByName**
 - Returns array of elements that have given name attribute
- **getElementsByTagName**
 - Returns array of elements that have given element name
 - Case insensitive
 - Inherited from Document class (see earlier slide)
- **getElementById (inherited from Document)**
 - Finds element with specified id attribute
 - Inherited from Document class (see earlier slide)

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HTMLElement

- **Subclass of Element class. Obtain with**
 - document.body, document.getElementsByTagName, document.getElementsByName, document.images. etc.
 - otherElement.getElementsByTagName, otherElement.childNodes, otherElement.firstChild, etc.
- **Most important properties**
 - id
 - The id attribute
 - nodeName
 - Element name (inherited from Node class).
 - name
 - The name attribute (for HTML elements with "name" only)
 - **innerHTML**
 - Read/write property giving HTML text inside element
 - style
 - CSS2Properties object representing element styling
 - className
 - Space-separated list of CSS class names
- **Method**
 - scrollToView
 - Scrolls browser so element is visible

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Form Class

- **Obtaining reference**
 - document.forms array, "form" variable for code invoked by input element
 - Any method or property that returns Node (getElementById, childNodes, etc.)
- **Properties**
 - elements
 - Array of all input elements in form
 - action, encType, method, name, target
 - Corresponds to HTML attributes
- **Methods**
 - submit, reset

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Image Class

- **Obtaining reference**
 - document.images array, document.nameOfImage
 - Any method or property that returns Node (getElementById, childNodes, etc.)
- **Properties**
 - src
 - Read/write property. Changing this changes the image.
 - Preload images first with new Image(...) and setting its href, so browser will have images cached.
 - name, align, alt, border, height, hspace, ismap, usemap, vspace, width
 - Corresponds to HTML attributes
 - Inherited ones described earlier
 - Node, Element, HTMLElement

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Input Class

- **Obtaining reference**
 - theForm.elements array
 - Any method or property that returns Node (getElementById, childNodes, etc.)
- **Properties**
 - name, id, value, type, disabled, form (enclosing form)
 - For all input elements
 - defaultValue
 - Initial value as given in the HTML
 - Type-specific properties (see online API)
 - checked, maxLength, useMap, etc.
- **Methods**
 - blur/focus (all), click (buttons, checkboxes, radio buttons), select (file, password, text)

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Window Class: Properties

- **Obtaining reference**
 - Use "window" or "self"
- **history**
 - History object. Not writable.
- **location**
 - Location object.
 - location.href = "new address" redirects browser
- **status**
 - Status line value. Writable.
- **Size/position/scrolling**
 - innerWidth, innerHeight, outerWidth, outerHeight, screenX (or screenLeft), screenY (or screenTop)

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Window Class: Methods

- **alert, confirm, prompt**
 - Pops up dialog box
- **print**
 - Invokes print dialog
- **setInterval, clearInterval [repeated actions]**
 - setInterval(someFunction, milliseconds)
- **setTimeout, clearTimeout [one-time actions]**
 - setTimeout(someFunction, milliseconds)
- **getComputedStyle**
 - Get style info for specified element
- **Movement**
 - Lots of methods for opening, closing, resizing windows

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Event Handling

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Idea

- **Most HTML elements have onXxx attributes**
 - Designates function to run when specified event occurs
- **General event handlers**
 - onclick, ondblclick
 - User single-clicks or double-clicks over element
 - onkeydown, onkeypress, onkeyup
 - User presses key when element has the focus
 - onmousedown, onmouseup
 - User presses mouse over element
 - onmouseover, onmouseout
 - Mouse moves over or leaves being over element
 - onmousemove
 - Mouse moves while over element

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Event Handlers: Example (HTML)

```
...
<head><title>General Event Handlers</title>
...
<script src="./scripts/general-events.js"
        type="text/javascript"></script>
</head>
<body>
...
<h2 onclick="alert('Ouch!')">
Here is a heading.
What happens when you click on it?
</h2>
<h2 onmouseover="on()" onmouseout="off()">
Here is a heading.
<span id="messageRegion"></span>
What happens when you move over it?
</h2>
...
```

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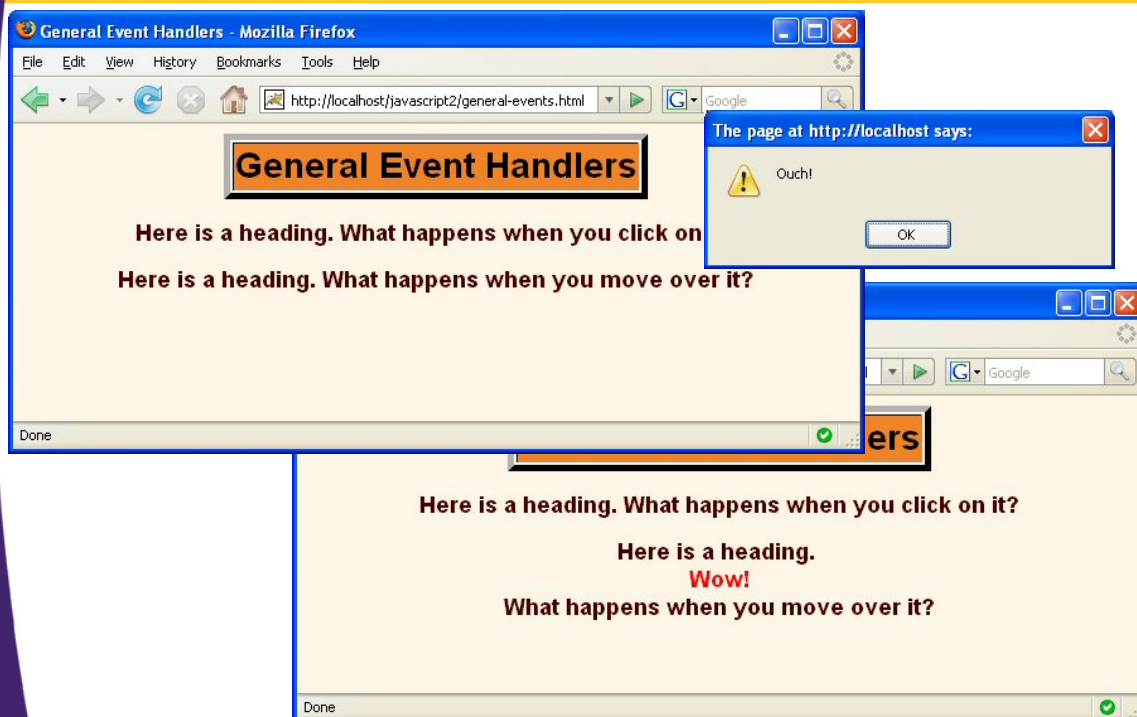
Event Handlers: Example (JavaScript)

```
function on() {  
    var span =  
        document.getElementById("messageRegion");  
    span.innerHTML =  
        "<br/><font color='red'>Wow!</font><br/>";  
}  
  
function off() {  
    var span =  
        document.getElementById("messageRegion");  
    span.innerHTML = "";  
}
```

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Event Handlers: Example (Results)



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Specialized Event Handlers

- **input**
 - onclick
 - For pushbuttons and toggle buttons. Also fires when button is invoked via keyboard.
 - onchange
 - For textfields, when change is committed (focus leaves)
 - Use onkeyup for individual characters
 - onblur, onfocus
- **form**
 - onsubmit, onreset
 - Return false to prevent form from really being submitted. Widely used for client-side validation of form fields.
- **body**
 - onblur, onerror, onfocus, onload, onresize, onunload
- **img**
 - onabort, onerror, onload

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Summary

- **XML**
 - Obtaining document
 - request.responseXML or parseFromString (testing only)
 - Node and Element properties and methods
 - attributes, childNodes, firstChild, nodeName, nodeValue
 - getElementsByTagName, getAttribute, normalize
- **HTML**
 - Methods and properties
 - document.writeln, getElementById
 - name, id, value, type, disabled, innerHTML
- **Event Handling**
 - `<input type="button" onclick="someFunction()"/>`

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Questions?

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